Reactions in plants

9B Plant growth · Lesson 1 of 7

Date _____ Class _____

Today I can 今天我能

- ☐ I can write the word and symbol equations for photosynthesis and aerobic respiration, and say where and when each happens in a plant.
- ☐ I can explain how limiting factors slow photosynthesis and why roots need oxygen.

What happens today

1. Do Now: on a farm
2. Learn: Photosynthesis · Light, chlorophyll and limiting factors · and more
3. Core tasks
4. Challenge, then the exit ticket

Student Book pages 21 to 23 · Workbook pages 15 to 16

What I learned today (one or two sentences) 今天我学到了：

Key Vocabulary

Term	中文	Definition
photosynthesis	光合作用	the process plants use to make glucose from carbon dioxide and water, using light
aerobic respiration	有氧呼吸	the reaction in every living cell that releases energy from glucose using oxygen
chlorophyll	叶绿素	the green substance that captures energy from light
chloroplast	叶绿体	the part of a plant cell where photosynthesis happens
glucose	葡萄糖	the sugar made in photosynthesis
reactant	反应物	a substance used up in a chemical reaction
product	产物	a substance made in a chemical reaction
limiting factor	限制因素	a variable that slows down the rate of photosynthesis
phloem	韧皮部	tubes made of living cells that carry sugar solution around a plant
yield	产量	the amount of useful product a farmer gets from a crop

Command words 指令词: Calculate 计算 · Complete 完成（填写） · Draw 画出 · Evaluate 评价 · Explain 解释 · Match 连线（配对） · Tick 打勾 · Use 使用（运用） · Write 写出

Quick Review

on a farm

Farmers want as much useful product (yield) from their crops as possible.

1 What process do wheat plants use to make their own food?

2 Name the gas that plants take in for this process.

3 Suggest the effect of a very cloudy year on the wheat harvest.

Core 1 of 4

Everyone does these. Answer on the card or under it. 人人都做，答案写在卡片上或下面。

1 Complete the word equations. Use the word bank.

Photosynthesis: _____ + water makes _____ + oxygen

Aerobic respiration: glucose + _____ makes carbon dioxide + _____ (+ energy released)

2 Tick one box in each row to show whether the statement is about photosynthesis or aerobic respiration.

Statement	Photosynthesis	Respiration
happens only when there is light		
happens all the time, day and night		
takes in carbon dioxide		
releases energy from glucose		
happens in chloroplasts		
happens in every living cell		

Core 2 of 4

Everyone does these. Answer on the card or under it. 人人都做，答案写在卡片上或下面。

3 Match each key word to its meaning. Draw a line.

chlorophyll	tubes of living cells that carry sugar solution around the plant
chloroplast	the sugar made in photosynthesis
limiting factor	the green substance that captures energy from light
phloem	a variable that slows down the rate of photosynthesis
glucose	the part of a plant cell where photosynthesis happens

Core 3 of 4

Everyone does these. Answer on the card or under it. 人人都做，答案写在卡片上或下面。

4 Van Helmont grew a willow tree in a pot of dry soil for five years. He added only rainwater. The table shows his results.

	At the start	After five years
mass of dry soil (kg)	90.72	90.66
mass of tree (kg)	2.27	76.74

a) Calculate the change in the mass of the soil.

b) Calculate the change in the mass of the tree.

Core 4 of 4

Everyone does these. Answer on the card or under it. 人人都做，答案写在卡片上或下面。

4 *continued*

c) Aristotle thought that roots sucked up soil for the plant to eat. Explain how these results show that Aristotle was wrong.

Challenge

When Core is finished. Full sentences. 完成基础题后做；请写完整句子。

C1 Draw a flow chart to show how energy from the Sun becomes stored in a plant. Use the words light, chlorophyll, chloroplast and glucose.

Draw here. 画在这里。

C2 After his experiment, van Helmont concluded that all the extra mass of the tree came from the water he added. Evaluate this conclusion. Use the word equation for photosynthesis in your answer.

Exit Ticket

Hand this card in at the end of the lesson.

Name _____

1 Write the word equation for photosynthesis.

2 Where in a plant cell does photosynthesis happen?

3 Why do plants in a field only photosynthesise during the day?

Today I can: Goal 1 ☐ Goal 2 ☐